

# VESSEL ANALYSIS REPORT

| vaf(%) | branch points count | tip point count | vein count | total vein length | average vein length |
|--------|---------------------|-----------------|------------|-------------------|---------------------|
| 1.534  | 56.0                | 146.0           | 158.0      | 4452.438          | 28.18               |

| id | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|----|------------|------------|--------|---------|---------|
| 1  | [218,3]    | [225,11]   | 11.071 | TIP     | BRANCH  |
| 2  | [225,11]   | [240,11]   | 14.0   | BRANCH  | BRANCH  |
| 3  | [225,11]   | [188,66]   | 77.598 | BRANCH  | BRANCH  |
| 4  | [250,56]   | [240,11]   | 48.971 | TIP     | BRANCH  |
| 5  | [253,3]    | [240,11]   | 14.899 | TIP     | BRANCH  |
| 6  | [128,14]   | [92,25]    | 39.556 | BRANCH  | BRANCH  |
| 7  | [144,29]   | [128,14]   | 21.799 | TIP     | BRANCH  |
| 8  | [128,14]   | [169,15]   | 52.385 | BRANCH  | BRANCH  |
| 9  | [196,19]   | [169,15]   | 29.071 | TIP     | BRANCH  |
| 10 | [169,15]   | [144,73]   | 67.355 | BRANCH  | BRANCH  |
| 11 | [277,1]    | [285,16]   | 17.899 | TIP     | BRANCH  |
| 12 | [298,24]   | [285,16]   | 15.314 | TIP     | BRANCH  |
| 13 | [285,16]   | [299,68]   | 62.941 | BRANCH  | BRANCH  |
| 14 | [92,25]    | [44,46]    | 56.284 | BRANCH  | BRANCH  |
| 15 | [92,25]    | [104,65]   | 44.799 | BRANCH  | BRANCH  |
| 16 | [485,30]   | [494,35]   | 10.657 | TIP     | BRANCH  |
| 17 | [509,35]   | [494,35]   | 14.828 | TIP     | BRANCH  |
| 18 | [492,57]   | [494,35]   | 22.243 | TIP     | BRANCH  |
| 19 | [44,46]    | [68,81]    | 44.527 | BRANCH  | BRANCH  |
| 20 | [44,46]    | [22,118]   | 84.012 | BRANCH  | BRANCH  |
| 21 | [104,65]   | [68,81]    | 41.213 | BRANCH  | BRANCH  |
| 22 | [112,72]   | [104,65]   | 9.485  | TIP     | BRANCH  |
| 23 | [195,68]   | [188,66]   | 6.828  | TIP     | BRANCH  |
| 24 | [188,66]   | [184,67]   | 3.0    | BRANCH  | BRANCH  |
| 25 | [179,71]   | [183,67]   | 5.243  | TIP     | BRANCH  |
| 26 | [183,67]   | [183,65]   | 1.414  | BRANCH  | PATH    |

| id | p1.x, p1.y | p2.x, p2.y | length  | p1_type | p2_type |
|----|------------|------------|---------|---------|---------|
| 27 | [299,68]   | [343,116]  | 71.083  | BRANCH  | BRANCH  |
| 28 | [299,68]   | [290,88]   | 22.314  | BRANCH  | BRANCH  |
| 29 | [138,69]   | [144,73]   | 6.657   | TIP     | BRANCH  |
| 30 | [144,73]   | [143,93]   | 19.414  | BRANCH  | BRANCH  |
| 31 | [476,44]   | [444,76]   | 52.456  | TIP     | BRANCH  |
| 32 | [388,122]  | [444,76]   | 75.811  | TIP     | BRANCH  |
| 33 | [450,89]   | [444,76]   | 14.071  | TIP     | BRANCH  |
| 34 | [68,81]    | [68,86]    | 4.0     | BRANCH  | BRANCH  |
| 35 | [127,83]   | [125,84]   | 1.414   | TIP     | BRANCH  |
| 36 | [124,87]   | [125,84]   | 2.414   | TIP     | BRANCH  |
| 37 | [120,79]   | [125,84]   | 5.657   | TIP     | BRANCH  |
| 38 | [68,86]    | [38,118]   | 44.012  | BRANCH  | BRANCH  |
| 39 | [68,86]    | [74,123]   | 43.042  | BRANCH  | BRANCH  |
| 40 | [275,98]   | [290,88]   | 18.142  | TIP     | BRANCH  |
| 41 | [296,97]   | [290,88]   | 10.071  | TIP     | BRANCH  |
| 42 | [143,93]   | [117,125]  | 44.698  | BRANCH  | BRANCH  |
| 43 | [152,100]  | [143,93]   | 10.485  | TIP     | BRANCH  |
| 44 | [232,42]   | [211,95]   | 62.113  | TIP     | BRANCH  |
| 45 | [211,95]   | [142,162]  | 116.74  | BRANCH  | BRANCH  |
| 46 | [212,97]   | [211,95]   | 1.414   | TIP     | BRANCH  |
| 47 | [511,82]   | [495,105]  | 28.627  | TIP     | BRANCH  |
| 48 | [468,113]  | [495,105]  | 30.485  | TIP     | BRANCH  |
| 49 | [499,122]  | [495,105]  | 18.071  | TIP     | BRANCH  |
| 50 | [370,280]  | [343,116]  | 188.125 | TIP     | BRANCH  |
| 51 | [340,117]  | [343,116]  | 2.414   | TIP     | BRANCH  |
| 52 | [22,118]   | [38,118]   | 15.0    | BRANCH  | BRANCH  |
| 53 | [22,118]   | [15,170]   | 54.314  | BRANCH  | BRANCH  |
| 54 | [38,118]   | [44,124]   | 8.071   | BRANCH  | BRANCH  |
| 55 | [74,123]   | [44,124]   | 29.414  | BRANCH  | BRANCH  |
| 56 | [74,123]   | [70,145]   | 22.657  | BRANCH  | BRANCH  |
| 57 | [44,124]   | [48,141]   | 17.657  | BRANCH  | BRANCH  |
| 58 | [117,125]  | [97,162]   | 47.213  | BRANCH  | BRANCH  |

| id | p1.x, p1.y | p2.x, p2.y | length  | p1_type | p2_type |
|----|------------|------------|---------|---------|---------|
| 59 | [117,125]  | [106,163]  | 46.213  | BRANCH  | BRANCH  |
| 60 | [48,141]   | [25,170]   | 38.113  | BRANCH  | BRANCH  |
| 61 | [48,141]   | [70,145]   | 22.243  | BRANCH  | BRANCH  |
| 62 | [70,145]   | [70,147]   | 1.0     | BRANCH  | BRANCH  |
| 63 | [70,147]   | [93,162]   | 28.799  | BRANCH  | BRANCH  |
| 64 | [64,188]   | [70,147]   | 43.071  | TIP     | BRANCH  |
| 65 | [93,162]   | [97,162]   | 3.0     | BRANCH  | BRANCH  |
| 66 | [93,162]   | [82,237]   | 86.426  | BRANCH  | BRANCH  |
| 67 | [97,162]   | [106,163]  | 8.0     | BRANCH  | BRANCH  |
| 68 | [135,161]  | [142,162]  | 6.414   | TIP     | BRANCH  |
| 69 | [145,183]  | [142,162]  | 21.243  | TIP     | BRANCH  |
| 70 | [106,170]  | [106,163]  | 6.414   | TIP     | BRANCH  |
| 71 | [15,170]   | [25,170]   | 9.0     | BRANCH  | BRANCH  |
| 72 | [15,170]   | [26,234]   | 77.426  | BRANCH  | BRANCH  |
| 73 | [25,170]   | [32,182]   | 13.485  | BRANCH  | BRANCH  |
| 74 | [32,182]   | [46,200]   | 26.899  | BRANCH  | BRANCH  |
| 75 | [26,194]   | [32,182]   | 13.071  | TIP     | BRANCH  |
| 76 | [55,197]   | [46,200]   | 9.243   | TIP     | BRANCH  |
| 77 | [46,200]   | [26,234]   | 47.728  | BRANCH  | BRANCH  |
| 78 | [196,180]  | [195,223]  | 44.899  | TIP     | BRANCH  |
| 79 | [190,223]  | [195,223]  | 4.0     | TIP     | BRANCH  |
| 80 | [207,249]  | [195,223]  | 29.971  | TIP     | BRANCH  |
| 81 | [26,234]   | [24,244]   | 9.828   | BRANCH  | BRANCH  |
| 82 | [82,237]   | [76,238]   | 5.414   | BRANCH  | BRANCH  |
| 83 | [82,237]   | [98,244]   | 18.485  | BRANCH  | BRANCH  |
| 84 | [76,238]   | [75,249]   | 10.414  | BRANCH  | BRANCH  |
| 85 | [57,224]   | [76,238]   | 24.556  | TIP     | BRANCH  |
| 86 | [24,244]   | [26,246]   | 133.368 | BRANCH  | PATH    |
| 87 | [108,242]  | [98,244]   | 9.828   | TIP     | BRANCH  |
| 88 | [96,252]   | [98,244]   | 7.828   | TIP     | BRANCH  |
| 89 | [57,244]   | [75,249]   | 19.071  | TIP     | BRANCH  |
| 90 | [75,249]   | [101,290]  | 52.113  | BRANCH  | BRANCH  |

| id  | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|-----|------------|------------|--------|---------|---------|
| 91  | [255,262]  | [244,282]  | 24.142 | TIP     | BRANCH  |
| 92  | [198,326]  | [244,282]  | 70.841 | TIP     | BRANCH  |
| 93  | [249,286]  | [244,282]  | 5.243  | TIP     | BRANCH  |
| 94  | [120,289]  | [101,290]  | 20.071 | TIP     | BRANCH  |
| 95  | [101,290]  | [83,311]   | 27.627 | BRANCH  | BRANCH  |
| 96  | [77,311]   | [83,311]   | 5.0    | TIP     | BRANCH  |
| 97  | [83,311]   | [95,347]   | 39.971 | BRANCH  | BRANCH  |
| 98  | [2,303]    | [5,346]    | 44.899 | TIP     | BRANCH  |
| 99  | [19,348]   | [5,346]    | 13.828 | TIP     | BRANCH  |
| 100 | [3,350]    | [5,346]    | 3.828  | TIP     | BRANCH  |
| 101 | [101,355]  | [95,347]   | 10.071 | TIP     | BRANCH  |
| 102 | [91,349]   | [95,347]   | 3.414  | TIP     | BRANCH  |
| 103 | [39,326]   | [67,353]   | 45.213 | TIP     | BRANCH  |
| 104 | [82,352]   | [67,353]   | 14.414 | TIP     | BRANCH  |
| 105 | [34,351]   | [67,353]   | 34.899 | TIP     | BRANCH  |
| 106 | [55,379]   | [47,394]   | 17.314 | TIP     | BRANCH  |
| 107 | [41,392]   | [47,394]   | 5.828  | TIP     | BRANCH  |
| 108 | [47,394]   | [60,439]   | 49.385 | BRANCH  | BRANCH  |
| 109 | [19,403]   | [16,431]   | 29.071 | TIP     | BRANCH  |
| 110 | [3,431]    | [16,431]   | 12.0   | TIP     | BRANCH  |
| 111 | [40,441]   | [16,431]   | 26.728 | TIP     | BRANCH  |
| 112 | [70,443]   | [60,439]   | 10.657 | TIP     | BRANCH  |
| 113 | [7,503]    | [60,439]   | 95.497 | TIP     | BRANCH  |
| 114 | [386,287]  | [362,463]  | 185.77 | TIP     | BRANCH  |
| 115 | [334,457]  | [362,463]  | 29.485 | TIP     | BRANCH  |
| 116 | [362,463]  | [362,476]  | 12.0   | BRANCH  | BRANCH  |
| 117 | [364,476]  | [362,476]  | 1.0    | TIP     | BRANCH  |
| 118 | [332,488]  | [362,476]  | 33.556 | TIP     | BRANCH  |
| 119 | [330,0]    | [317,32]   | 40.314 | TIP     | TIP     |
| 120 | [391,0]    | [364,5]    | 30.728 | TIP     | TIP     |
| 121 | [405,2]    | [406,12]   | 10.414 | TIP     | TIP     |
| 122 | [463,2]    | [422,11]   | 50.971 | TIP     | TIP     |

| id  | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|-----|------------|------------|--------|---------|---------|
| 123 | [378,24]   | [381,57]   | 35.071 | TIP     | TIP     |
| 124 | [134,51]   | [135,56]   | 6.243  | TIP     | TIP     |
| 125 | [342,58]   | [336,70]   | 14.485 | TIP     | TIP     |
| 126 | [215,106]  | [215,112]  | 6.0    | TIP     | TIP     |
| 127 | [195,111]  | [191,112]  | 4.414  | TIP     | TIP     |
| 128 | [305,113]  | [317,139]  | 32.142 | TIP     | TIP     |
| 129 | [271,114]  | [256,154]  | 46.213 | TIP     | TIP     |
| 130 | [211,137]  | [228,145]  | 21.142 | TIP     | TIP     |
| 131 | [299,142]  | [288,144]  | 12.657 | TIP     | TIP     |
| 132 | [457,142]  | [462,145]  | 6.243  | TIP     | TIP     |
| 133 | [499,147]  | [486,170]  | 28.385 | TIP     | TIP     |
| 134 | [433,156]  | [423,161]  | 12.071 | TIP     | TIP     |
| 135 | [322,157]  | [327,160]  | 7.657  | TIP     | TIP     |
| 136 | [308,190]  | [315,200]  | 13.485 | TIP     | TIP     |
| 137 | [221,197]  | [219,201]  | 4.828  | TIP     | TIP     |
| 138 | [164,205]  | [167,214]  | 11.071 | TIP     | TIP     |
| 139 | [422,210]  | [438,224]  | 22.385 | TIP     | TIP     |
| 140 | [444,228]  | [455,232]  | 12.657 | TIP     | TIP     |
| 141 | [144,252]  | [145,253]  | 1.414  | TIP     | TIP     |
| 142 | [300,276]  | [317,283]  | 27.314 | TIP     | TIP     |
| 143 | [222,327]  | [224,331]  | 4.828  | TIP     | TIP     |
| 144 | [140,333]  | [130,343]  | 14.728 | TIP     | TIP     |
| 145 | [166,351]  | [160,365]  | 16.485 | TIP     | TIP     |
| 146 | [4,374]    | [23,390]   | 26.799 | TIP     | TIP     |
| 147 | [70,387]   | [87,390]   | 18.243 | TIP     | TIP     |
| 148 | [128,396]  | [148,418]  | 32.627 | TIP     | TIP     |
| 149 | [107,402]  | [117,407]  | 12.071 | TIP     | TIP     |
| 150 | [449,403]  | [450,407]  | 4.414  | TIP     | TIP     |
| 151 | [79,404]   | [75,408]   | 5.657  | TIP     | TIP     |
| 152 | [184,424]  | [165,478]  | 63.284 | TIP     | TIP     |
| 153 | [510,438]  | [464,451]  | 51.385 | TIP     | TIP     |
| 154 | [431,443]  | [417,447]  | 15.657 | TIP     | TIP     |

| id  | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|-----|------------|------------|--------|---------|---------|
| 155 | [312,447]  | [311,455]  | 9.243  | TIP     | TIP     |
| 156 | [97,454]   | [108,472]  | 23.142 | TIP     | TIP     |
| 157 | [120,485]  | [132,508]  | 29.385 | TIP     | TIP     |
| 158 | [105,490]  | [104,501]  | 11.414 | TIP     | TIP     |